7-1 Project

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Purpose and Specific Problems

The accounting department is comparing Microsoft Excel Power Query/Power BI and Tableau to determine which software product is more beneficial for recurring accounting analytics. The company needs a tool that can prepare data, identify anomalies, summarize large transaction populations, and communicate conclusions to both technical and non-technical audiences. The labs tested several common accounting problems: cleaning inconsistent Lending Club fields, summarizing more than 235,000 loan records, identifying borrower clusters and outliers, finding duplicate payments in audit data, analyzing job cost components, evaluating cost variances, and creating balanced-scorecard KPIs.

Both platforms solved important problems, but they solved them differently. Power Query/Power BI was strongest when the work required cleaning, modeling, audit traceability, duplicate-payment logic, DAX measures, and integration with the Microsoft environment. Tableau was strongest when the work required quick visual exploration, drag-and-drop dashboard design, filters, color-coded performance comparisons, and non-technical communication. Because accounting workflows require both reliable preparation and defensible results, the final recommendation is to adopt Microsoft Excel Power Query/Power BI as the primary tool and use Tableau as a supplemental visualization option when highly polished dashboards are needed.

Techniques, Clean Data, and Ethical Issues

Area	Microsoft Excel Power Query/Power BI	Tableau
Preparation and cleaning	Power Query replaced	Tableau Prep can shape data,

	inconsistent values, profiled columns, standardized types, and revealed errors or blanks before analysis.	but Tableau Public/Desktop visual workflows were less natural for cleaning than Power Query in these course tasks.
Analysis techniques	Summary statistics, KPI cards, slicers, DAX measures, duplicate flags, variance calculations, and scorecard thresholds.	Rows/Columns shelves, filters, calculated fields, cluster analysis, parameters, maps, stacked bars, and dashboards.
Communication	Power BI dashboards connected clean data to drill-down reports and Microsoft-friendly reporting.	Tableau produced fast, visually clear dashboards for product margin, delivery time, salespeople, and customers.

Clean data matters because errors at the preparation stage flow into every statistic, chart, and recommendation. In the Lending Club data, employment length, income, interest rate, and loan amount fields needed consistent formats before aggregation. In accounting, unclean data can distort averages, hide exceptions, and weaken audit evidence. Ethical issues also affected the comparison. Lending and audit data can contain sensitive financial information, so privacy, limited access, and confidential handling are required. Bias must also be considered because historical lending data may reflect unequal patterns in borrower characteristics. Finally, the analyst must avoid misleading reporting by showing outliers, limitations, and unfavorable findings rather than selecting only positive visuals.

Anomalies, Quantitative Methods, and Problem Resolution

The analysis identified several anomalies and problem areas. First, Power Query/Power BI surfaced extreme values, including an annual income maximum that was far above the median borrower income; such values can distort averages and should be investigated before management relies on the result. Second, cluster visuals in Power BI and Tableau showed unusual borrower groups by debt-to-income ratio, loan amount, and interest rate. Third, the duplicate-payment lab identified repeated payment characteristics by payment ID, date, preparer, and invoice reference. Fourth, job-cost and variance dashboards showed where actual material, labor, overhead, and profit results differed from targets.

The quantitative methods used across the tools included descriptive statistics, aggregation, median and count measures, cluster analysis, variance analysis, ratio analysis, KPI thresholding, and dashboard filtering. Power BI supported these methods through Power Query transformations, DAX measures, slicers, KPI cards, and dashboards. Tableau supported the same decision process with calculated fields, clusters, color-coded variance visuals, parameters, and interactive dashboards. The methods used to resolve problems included standardizing inconsistent fields, filtering and drilling into exceptions, building duplicate flags, comparing actual values to targets, and tracking KPIs after management action.

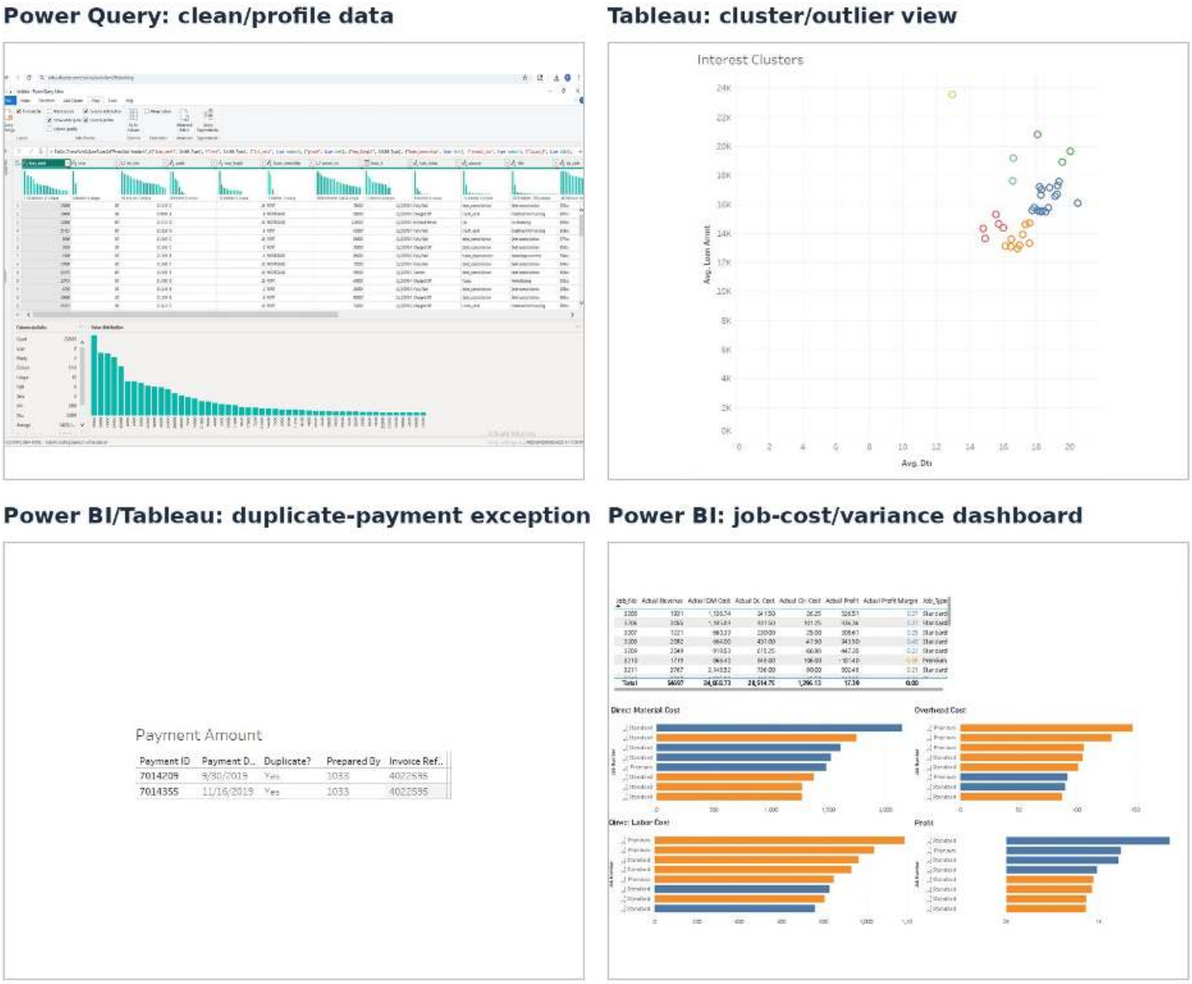

Figure 1. Evidence from the course labs: Power Query profiling, Tableau clustering, duplicate-payment detection, and job-cost/variance visuals.

Advantages, Disadvantages, and Recommendation

Power BI advantages were the integrated cleaning workflow, strong compatibility with Excel, measurable audit logic, and the ability to combine preparation, modeling, and dashboards in one Microsoft ecosystem. The disadvantages were the learning curve for DAX, crowded interfaces, repetitive setup steps, and layout complexity when building dashboards. Tableau advantages were speed, drag-and-drop visual exploration, clean dashboards, strong cluster visuals, maps, and easy non-technical communication. Tableau disadvantages were that the public publishing workflow is not suitable for confidential accounting data, some calculated-field and relationship steps were initially confusing, and deeper data preparation required Tableau Prep rather than the basic visual interface.

The recommended software option is Microsoft Excel Power Query/Power BI. The reason is not that Tableau performed poorly; Tableau was often faster and easier for visual storytelling. However, the accounting department needs a primary tool that supports repeatable data preparation, audit analytics, anomaly review, duplicate-payment testing, variance analysis, KPI monitoring, and controlled reporting. Power Query/Power BI best fits that full workflow, while Tableau is best positioned as a supplemental visualization tool for leadership-facing dashboards.

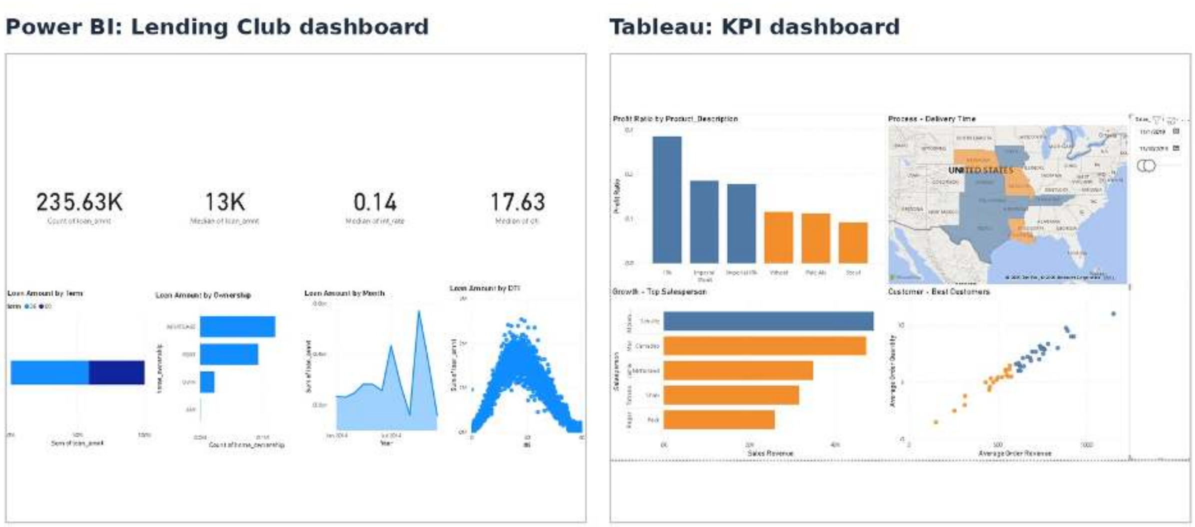

Figure 2. Dashboard evidence comparing Power BI lending summaries with Tableau KPI visual storytelling.

Leadership Summary

For leadership, the data story should be simple: both tools can turn accounting data into useful insights, but Power BI provides a stronger end-to-end accounting analytics environment. Tableau is useful when the primary need is fast, visually appealing exploration. The department should implement Power BI for routine accounting analytics and dashboards, establish data-quality checks before every report, use anomaly and duplicate-payment rules as recurring controls, and reserve Tableau for selected presentations where visual simplicity and executive communication are the main goals.

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